

Zheng Zixuan

Gender: Male Objective: Backend Engineer Location: Shanghai

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Education

2013.09 - 2017.06 Shandong University Network Engineering (B.S.)

Courses: Operating Systems, Design Patterns, Computer Organization, Machine Learning, Distributed Systems, Data Structures and Algorithms

Summary

Backend engineer with several years of Java experience and an ~1-year career gap, self-studied and contributed to open source during the break.

Work Experience

2021.04 - Present Zhihu Backend Engineer

- Owned backend of core transaction and order services, splitting the monolith into order, settlement and risk-control microservices, with experience documented and shared internally.
- Tuned Redis caching for hot keys and cache breakdown with local cache and distributed locks, raising hit rate above 90%, and it became a reusable team best practice.
- Designed a Kafka-based async transaction pipeline with idempotency keys, retry queues and dead-letter handling, sustaining 20000 QPS at peak, and it became a reusable team best practice.

Career gap: 2020.05 - 2021.04 (break for personal reasons; self-studied and contributed to open source)

2017.07 - 2020.05 Tencent Backend Engineer

- Fixed a log-collection failure caused by agent backpressure, restoring stability after tuning batch and buffer settings.
- Led a production incident review with root-cause analysis and drove an alert-tiering and on-call rotation mechanism.
- Rebuilt the config and template management service to support hot template reload without restart, cutting release time by 35%, and key metrics improved steadily.

Projects

High-Concurrency Flash-Sale & Inventory System (Java + Redis + RocketMQ)

- Built circuit breaking and fallback for downstream failures, ensuring online service stability.
- Pre-deducted inventory in Redis with async DB writes to absorb traffic spikes.
- Stayed stable at 20000 QPS in load tests with error rate below 0.1%, ensuring online service stability.
- Designed distributed locks and token-bucket rate limiting to prevent oversell.

Enterprise Auth & Gateway Platform (Spring Cloud Gateway + OAuth2)

- Integrated OAuth2 and RBAC with multi-tenant isolation, significantly lowering maintenance cost.
- Shipped a unified SDK to lower integration cost across teams.
- Unified auth, rate limiting, gray routing and tracing as reusable platform capabilities, and it became a reusable team best practice.
- Added circuit breaking to isolate faulty downstreams, improving end-to-end delivery quality.

Distributed Payment & Settlement Platform (Spring Boot + MySQL + Kafka + Redis)

- Recorded an audit trace id for every payment state transition for compliance.
- Used sharding and read-write splitting to keep single-table rows within tens of millions.
- Applied idempotency and eventual consistency to avoid duplicate charges on retry.
- Implemented daily reconciliation across payment gateway, ledger and settlement files with auto-correction.

Highlights

- Established metrics and reviews around root-cause analysis and postmortems to keep improving delivery quality
- Owned core modules that steadily served 5810K daily visits with 99.9%+ availability
- Built a reusable methodology and docs around Git collaboration and code review, adopted by several teams
- Continuously optimized CI/CD pipelines, cutting key-path latency from 2200ms to 320ms
- Led a major technical effort involving MySQL indexing and slow-query tuning, staying stable at 20000 QPS peak

Skills

Core skills: Distributed transactions and consistency, JVM tuning and troubleshooting, Backend

Proficient in: MySQL, Spring, Java

Familiar with: Docker and containerization, Redis caching and distributed locks, Kafka/RocketMQ messaging, MySQL indexing and slow-query tuning

Self Evaluation

- Enjoys retrospectives, cares about code quality and observability, and makes data-driven decisions
- Strong communication and collaboration with product, QA and ops to deliver projects on time